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The Editor-in-Chief
Electric Power Systems Research
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Dear Editor,

We would like to submit our manuscript, “Trigger-Based Variable-Fidelity Co-Simulation Scheduling for Reinforcement-Learning-Controlled Power-Electronic Systems: What Rigorous Validation Catches,” for consideration as an original research article in *Electric Power Systems Research*.

Relevance to the journal. The paper addresses a practical bottleneck in RL-based power-system control: training a voltage-regulation controller for switched capacitor banks on a distribution feeder requires a circuit-level simulator, and the fidelity of that simulator is normally fixed for the duration of the study. We present an online scheduling algorithm that switches the simulation fidelity at each control step based on a provable Lipschitz-derived error bound, a state rate-of-change trigger, and the RL agent’s action-entropy, keeping simulation cost low when the system is quiet and escalating automatically before significant transients. The algorithm provides a formal, computable certificate bounding how far the active low-fidelity trajectory may have drifted from the high-fidelity reference, which distinguishes it from simpler surrogate-error thresholds used in the multi-fidelity literature.

What this paper discloses. We want to flag, before referees encounter it, that the paper reports three substantive problems we found and corrected in our own original design rather than silently fixing them:

1. The conventional low-fidelity power-flow approximation (flat-start DC power flow) has no mechanism for voltage-magnitude response to reactive power – exactly what switched capacitor banks produce – and carried a mean tracking error of 0.09 p.u., larger than the full ANSI C84.1 voltage band. We replace it with a warm-started fast-decoupled power flow and quantify the improvement directly.
2. Profiling the medium-fidelity Newton–Raphson solver showed that Jacobian assembly, not the linear solve, dominates its cost; a cached-Jacobian variant cuts cost by roughly $3\times$ at negligible accuracy loss, giving the scheduler a genuinely useful intermediate tier.
3. Validating the error-bound theorem against a shadow high-fidelity reference – rather than checking the scheduler’s own internal trigger bookkeeping, which is what an earlier version of this work reported as a “0% violation rate” – found the bound was violated on 73–80% of low-fidelity steps, traced to a model-gap constant miscalibrated by roughly 450 times. Correcting it brings the validated violation rate to approximately 0.26%.

Key results. With all three corrections in place, evaluated over 10 random seeds with 95% confidence intervals, paired t -tests, and Cohen’s d effect sizes, the corrected scheduler achieves 1.3–

1.5 $\times$ compute speedup over always-high fidelity (translating to approximately 30% reduction in RL training wall-clock time) at a 6–12% accuracy cost relative to fixed periodic switching. The accuracy difference from periodic switching is not statistically significant ($p > 0.35$, $d < 0.31$) at $N = 10$ seeds; periodic’s speedup advantage is ($p < 0.02$, $d > 0.93$). We report this honestly alongside two advantages the trigger scheduler offers that periodic cannot: a real-time formal divergence certificate and a continuously tunable accuracy–speed operating point. The paper also includes a trigger-signal ablation, a threshold sweep, a sensitivity analysis, and a comparison against two independently designed baseline schedulers built on different mechanism classes.

This manuscript is original, has not been published previously, and is not under consideration elsewhere. All authors have approved the manuscript and agree with its submission to *Electric Power Systems Research*. We have no conflicts of interest to disclose. All code, data, and configuration required to reproduce every figure and table are available from the corresponding author upon reasonable request.

Thank you for your time and consideration.

Sincerely,

Md Hasibuzzaman
Chan-Yun Yang